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Quantifying Plyometric Intensity via Rate of Force Development, Knee Joint, and Ground Reaction Forces

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Abstract. Jensen, R.L., and W.P. Ebben. Quantifying plyometric intensity via rate of force development, knee joint, and ground reaction forces. *J. Strength Cond. Res. 21(3):763–767. 2007.*—Because the intensity of plyometric exercises usually is based simply upon anecdotal recommendations rather than empirical evidence, this study sought to quantify a variety of these exercises based on forces placed upon the knee. Six National Collegiate Athletic Association Division I athletes who routinely trained with plyometric exercises performed depth jumps from 46 and 61 cm, a pike jump, tuck jump, single-leg jump, countermovement jump, squat jump, and a squat jump holding dumbbells equal to 30% of 1 repetition maximum (RM). Ground reaction forces obtained via an AMTI force plate and video analysis of markers placed on the left hip, knee, lateral malleolus, and fifth metatarsal were used to estimate rate of eccentric force development (E-RFD), peak ground reaction forces (GRF), ground reaction forces relative to body weight (GRF/BW), knee joint reaction forces (K-JRF), and knee joint reaction forces relative to body weight (K-JRF/BW) for each plyometric exercise. One-way repeated measures analysis of variance indicated that E-RFD, K-JRF, and K-JRF/BW were different across the conditions ($p < 0.05$), but peak GRF and GRF/BW were not ($p > 0.05$). Results indicate that there are quantitative differences between plyometric exercises in the rate of force development during landing and the forces placed on the knee, though peak GRF forces associated with landing may not differ.

Key Words. stretch-shortening cycle, landing intensity, program design variables

Introduction

Plyometric exercises are widely used to develop muscular power, to enhance athletic performance, and to prevent injury. Like other training formats, plyometric program design requires an understanding of a variety of program design variables such as exercise mode, frequency, volume, program length, recovery, progression, and intensity (6). Intensity may be the most important of these variables. Typically, factors such as the number of points of contact during landing, the speed of the drill, the height of the jump, and the athlete's weight have been suggested as possible determinants of plyometric intensity (6). Similarly, anecdotal recommendations exist for categories of low- to high-intensity plyometric exercises (6).

The intensity of some conditioning modes, such as resistance training, is quantified easily because weight-training devices usually have clearly labeled masses and programs are frequently created at some percentage of an athlete's repetition maximum (RM) or 1RM. According to Ebben et al. (2), quantifying plyometric-exercise intensity requires consideration of both the mass of the mode of resistance or athlete and the acceleration of that mass. Adaptation to plyometric training may rely on attaining an intensity threshold, which has been described previously as the minimal essential eccentric strain (2). "Plyometric intensity" also has been defined as the amount of stress placed on involved muscles, connective tissues, and joints and is dictated by the type of plyometric exercise that is performed (6). Given this definition, it is plausible that plyometric intensity could be evaluated scientifically by assessing the stresses placed on muscle, connective tissue, and joints through variables such as the rate of force development (RFD), ground reaction force (GRF), and joint reaction force (JRF).

Previous research has examined the GRF and JRF of select plyometric exercises. Studies have compared kinetic and kinematic variables of drop jumps and pendulum jumps (3), as well as the GRF of unloaded and loaded drop jumping (8), of drop jumps of varying heights (7), and of one-legged and two-legged countermovement jumps (9). Research quantifying the intensity of a large number of plyometric exercises is limited to a study by Jensen and Ebben (4), who examined 10 different plyometric exercises and demonstrated that the impulse and GRF of plyometric exercises vary, depending on the type of exercise performed. Therefore, the purpose of the current study was to examine further the rate of eccentric force development (E-RFD), peak GRF, ground reaction forces relative to body weight (GRF/BW), knee joint reaction forces (K-JRF), and knee joint reaction forces relative to body weight (K-JRF/BW) of a variety of plyometric exercises in order to further quantify their intensity.

Methods

Experimental Approach to the Problem

A randomized repeated measures experimental design was used to test the hypothesis that differences existed among plyometric exercises. The independent variables used in this study included the 8 different plyometric exercises assessed. The dependent variables included the E-RFD, GRF, GRF/BW, K-JRF, and K-JRF/BW associated with each plyometric exercise.

Subjects

Six National Collegiate Athletic Association Division I athletes in track and field, volleyball, and wrestling (4 women and 2 men; mean ± *SD*; age = 20.3 ± 1.0 years, body mass = 78.9 ± 12.2 kg, 1RM squat = 143.6 ± 55.6 kg) volunteered to serve as subjects for the study. All subjects used the plyometric exercises evaluated in this study as a part of their regular strength training regimen; the participants included all-conference and all-American performers. Subjects completed a Physical Activity Readiness Questionnaire and gave informed consent prior to participating in the study. Approval for the use of human subjects was obtained from the university's Office of Research Compliance prior to commencing the study. Subjects had performed no strength training in the 48 hours prior to data collection.

Testing Procedures

Warm-up prior to the plyometric exercises consisted of at least 3 minutes of low-intensity work on a cycle ergometer. Warm-up was followed by static stretching, including 1 exercise for each major muscle group with stretches held from 12–15 seconds, and an activity-specific dynamic warm-up. Following the warm-up, subjects were allowed at least 5 minutes of rest prior to testing with the plyometric exercises. A variety of plyometric exercises, as well as exercises previously described and maximum power training (5), were included. The plyometric exercises were ordered randomly for each subject and consisted of depth jumps from 46 cm (DJ46) and 61 cm (DJ61), pike jump (PJ), tuck jump (TJ), single (left)-leg jump (SLJ), countermovement jump with arm swing (CMJ), squat jump with hands on top of the head (SJ), and a squat jump holding dumbbells equal to 30% of 1RM (SJ30). A 1-minute rest interval was maintained between each exercise.

Instrumentation

The plyometric exercises were performed by taking off from and landing on a 2-cm thick aluminum plate (76 × 102 cm) bolted directly to a force platform (OR6-5-2000; Advanced Mechanical Technology, Inc. [AMTI], Watertown, MA). The force platform configured with the oversized plate resulted in a natural frequency of not less than 142 Hz, within limits recommended by the manufacturer (AMTI) for this type of data collection. Ground reaction force data were collected at 1,000 Hz, real time displayed and saved with the use of computer software (BioSoft 1.0; AMTI) for later analysis. Peak GRF occurred during the plyometric landings and was defined as the highest value attained. Prior testing of this procedure indicated that reliability estimates for force measurement revealed an intraclass correlation coefficient of $R = 0.98$ (95% confidence interval = 0.94, −0.99). In addition, repeated measures analysis of variance (ANOVA) indicated there was no significant difference ($p > 0.05$) among 3 trials. The E-RFD was defined as the first peak of GRF divided by the time from onset of landing force to the first peak of GRF (1).

Video of the exercises was obtained at 60 Hz from the left side using 1-cm reflective markers placed on the greater trochanter, knee joint center, lateral malleolus, and the fifth metatarsal. Markers were digitized using Motus 5.2 (Peak Performance Technologies, Inc., Englewood, CO), and acceleration of the joint segment center of mass was determined after data was smoothed using a fourth-order Butterworth filter (11).

To synchronize data, a signal was used to initialize kinetic data collection, which also displayed a light in the view of the camera. Data then were combined into a single file and were splined to create a file of equal length at 60 Hz. Knee joint reaction forces were estimated according to methods previously described by Bauer et al. (1) using the equations illustrated in the Appendix. Because GRF for all but the SLJ would have been distributed across both feet (and therefore both knee joints),

Table 1. Peak ground reaction force (GRF), peak GRF relative to body weight (GRF/BW), and average rate of loading (mean ± *SD*) for 8 variations of plyometric exercises ($n = 6$).*

	DJ46	DJ61	CMJ	SJ	SJ30	SLJ	Pike	Tuck
GRF (N)	2,570 ± 979	3,056 ± 960	2,879 ± 892	2,389 ± 437	2,966 ± 1,112	2,261 ± 697	2,772 ± 767	2,937 ± 1,409
GRF/BW ($N \cdot kg^{-1}$)	3.25 ± 0.74	3.91 ± 0.80	3.71 ± 0.95	3.16 ± 0.84	3.74 ± 0.77	2.92 ± 0.81	3.70 ± 1.18	3.66 ± 1.33

* DJ46 = depth jump from 46 cm; DJ61 = depth jump from 61 cm; CMJ = countermovement jump; SJ = squat jump; SJ30 = squat jump holding dumbbells equal to 30% of 1 repetition maximum; SLJ = single-legged jump.

TABLE 2. Average rate of eccentric force development (mean ± *SD*) for 8 variations of plyometric exercises ($n = 6$).*

	DJ46	DJ61	CMJ	SJ	SJ30	SLJ	Pike	Tuck
E-RFD ($N\cdot s^{-1}$)	741 ± 347	975 ± 313	843 ± 357	505† ± 226	424† ± 168	718 ± 293	651 ± 232	804 ± 224

* DJ46 = depth jump from 46 cm; DJ61 = depth jump from 61 cm; CMJ = countermovement jump; SJ = squat jump; SJ30 = squat jump holding dumbbells equal to 30% of 1 repetition maximum; SLJ = single-legged jump; E-RFD = eccentric rate of force development.

† Significantly different ($p < 0.05$) from DJ46, DJ61, CMJ, SLJ, Tuck.

GRF values for all but the SLJ were divided by 2 prior to calculation of K-JRF.

Statistical Analyses

Statistical treatment of the data was performed using a 1-way (type of plyometric exercise) repeated measures analysis of variance (SPSS, version 11.5; SPSS, Inc., Chicago, IL) for average E-RFD, GRF, GRF/BW, K-JRF, and K-JRF/BW. An alpha level of 0.05 was used for all statistical tests. Effect size for each of the comparisons was estimated using the eta-squared statistic to describe the proportion of total variability attributable to an independent variable.

RESULTS

Peak GRF and GRF/BW were not statistically different ($p > 0.05$) across the 8 conditions (Table 1). In fact, the amount of variability demonstrated by the different exercises explained only 17.9 and 17.0% of the total variability for peak GRF and GRF/BW, respectively. However, differences were found in the E-RFD as well as in the K-JRF and K-JRF/BW. The E-RFD was lower ($p < 0.05$) for SJ and SJ30 than the DJ, CMJ, SLJ, and TJ were (Table 2). In this case, the exercise condition explained 50.3% of total variability according to the estimate of effect size. Similarly, peak K-JRF and K-JRF/BW also varied across the different types of jumps ($p < 0.05$), as shown in Table 3. The effect sizes for these peak K-JRF and K-JRF/BW were estimated to explain 56.1 and 53.8% of the total variability, respectively.

DISCUSSION

Peak GRF during plyometric exercises occurred during the landing portion of the exercises, as noted by previous authors (4, 6). Peak GRF and GRF/BW were not statistically different ($p > 0.05$) across the 8 conditions (Table 1). This was despite a wide range of body masses (66–96 kg). This finding is in contrast to previous work by Tsarouchas et al. (8), who found differences between loaded and unloaded DJ. Previous studies have demonstrated differences in GRF and impulse between plyometric variations (3, 4). In the present study, it is possible that the difference in findings were due to lack of statistical power as a result of the number of subjects and the relatively large variability in plyometric performance. However, differences were found in the E-RFD up to the point of the first peak GRF associated with variations of plyometric exercises. The first peak GRF (Figure 1) has been described previously as the contact of the toe followed by contact of the heel at peak 2 (1). In the present study, the E-RFD was lower ($p < 0.05$) for the SJ and SJ30 than for the DJ, CMJ, SLJ, and TJ (Table 2).

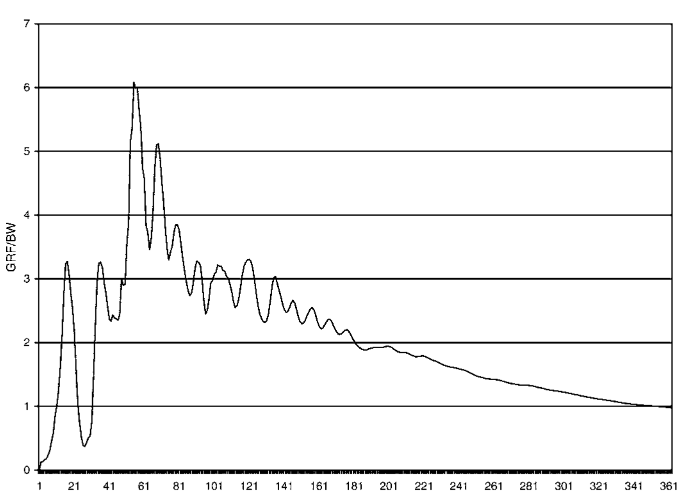

FIGURE 1. Ground reaction force relative to body mass for subject 3 during the landing of a drop jump from 61 cm.

These differences in the E-RFD across the jumping conditions appear to indicate variability among plyometric exercises and thus differences in their relative intensity. These differences in E-RFD are perhaps a function of landing techniques, which likely are influenced by the nature of the plyometric exercise itself, including factors such as jump height, unilateral vs. bilateral landing, added external load, and the degree to which forceful hip and knee extension is required in order for subjects to prepare their legs for landing (e.g., with the TJ and PJ). Researchers have suggested that a threshold E-RFD is necessary to optimally activate the stretch-shortening cycle (2). Although the best means to determine the threshold remains an elusive concept, the current data may be useful in determining a progression of plyometric exercises according to increasing E-RFD.

In addition to the E-RFD differences among plyometric exercises, peak K-JRF and K-JRF/BW also varied ($p < 0.05$), as shown in Table 3. Despite the small sample size of the current study ($n = 6$), the moderately large effect size for these variables appears to support the findings of significant differences among these exercises. Variations in K-JRF and K-JRF/BW, associated with different plyometric exercises, suggest that the relative intensity of these plyometric exercises can be determined empirically to some degree through these variables as well. Of particular interest is the variability in forces on the knee relative to body weight, with some plyometric exercises such as the PJ and TJ producing K-JRF nearly 10 times body weight, likely due to the subjects' active concentric activation of the knee and hip extensors to get the feet in position for landing after the large degree of hip flexion associated with the initial phase of these exercises. The SLJ also exhibited high K-JRF/BW values, which likely were due to the fact that all of the reaction force was distributed unilaterally and unilateral jump heights are typically approximately 58% of the bilateral equivalent (9). Exercises with added load resulted in lower jump heights and, therefore, somewhat lower K-JRF and K-JRF/BW. This was because the acceleration due to

TABLE 3. Peak knee reaction force (KRF) and peak KRF relative to body weight (KRF/BW) (mean ± *SD*) for 8 variations of plyometric exercises ($n = 6$).*

	DJ46	DJ61	CMJ	SJ	SJ30	SLJ	Pike	Tuck
KRF (N)	4,694‡ ± 1,348	4,620 ± 1,133	3,974† ± 1,265	5,232 ± 3,701	3,616‡ ± 948	6,638 ± 1,863	7,489 ± 1,822	7,566 ± 1,947
KRF/BW ($N \cdot kg^{-1}$)	5.98§ ± 1.26	5.97 ± 1.14	5.11† ± 1.21	7.07 ± 5.63	4.80‡ ± 1.33	8.54 ± 1.63	9.68 ± 1.95	9.94 ± 2.42

* DJ46 = depth jump from 46 cm; DJ61 = depth jump from 61 cm; CMJ = countermovement jump; SJ = squat jump; SJ30 = squat jump holding dumbbells equal to 30% of 1 repetition maximum; SLJ = single-legged jump.
† Significantly different ($p < 0.05$) from Pike, Tuck, SLJ.
‡ Significantly different ($p < 0.05$) from Pike, Tuck.
§ Significantly different ($p < 0.05$) from Pike, SLJ.

gravity was less, suggesting that these loaded plyometrics and maximum power training may be only moderately intense. The higher K-JRF and K-JRF/BW associated with the TJ, PJ, and SLJ suggest that these exercises should be introduced later in the development of a plyometric program as a way to increase plyometric intensity, particularly with those who are untrained or who are in the early stages of recovery from a knee injury. In addition to potential application with athletes, previous research with nonathletic populations has recommended that intensity variables such as GRF were useful to consider in the rehabilitation of clinical populations (10).

PRACTICAL APPLICATIONS

Quantifying the intensity of resistance training with machines or free weights is relatively easy and is determined in part by the loads that typically are labeled on machines or weight plates. For plyometrics, the best way to understand exercise intensity may be to quantify variables such as E-RFD, GRF, GRF/BW, K-JRF, and K-JRF/BW. Enhanced understanding of these variables of intensity is important to allow practitioners to progress plyometric intensity from low to high over the course of a program, consistent with recommendations that have been made for periodization of other forms of resistance training. This paper provides the coach and athlete with means of quantifying plyometric exercises in addition to the athlete's weight and the height of the jump, by combining these and other variables into the examination of forces placed upon the athlete.

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APPENDIX

The equations for calculating joint reaction forces are illustrated below (1).

$$R_x = -F_x + m_f \cdot a_{fx} + m_s \times a_{sx}$$

$$R_y = W_f + W_f - F_y + m_f \cdot a_{fy} + m_s \times a_{sy}$$

$$R = \sqrt{R_x + R_y}$$

where the variables in the equations are defined as follows:

R_x = reaction force of the knee in the horizontal direction;
R_y = reaction force of the knee in the vertical direction;
m_f = acceleration of the foot segment;
m_s = acceleration of the shank segment;
a_{fx} = acceleration of the foot segment center of mass; and
a_{sx} = acceleration of the shank segment center of mass.

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